

Support Vector Machines

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Introduction

- ▶ Consider a binary classification task
 - ▶ Support Vector Machines (SVM) try to find a plane that separates the two classes in the space of our predictive features
- ▶ If no such plane exists, there are two possible solutions
 - ▶ Relaxing what we mean by “separate”
 - ▶ Expanding our feature space to facilitate separation

Hyperplanes

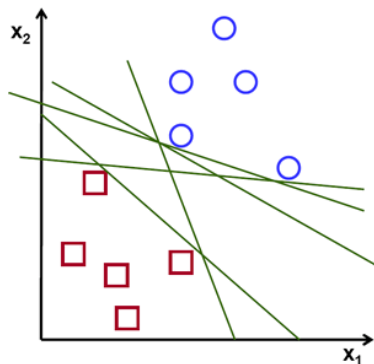
- ▶ A hyperplane is defined by a set of coefficients:

$$f(\mathbf{X}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

- ▶ Recognize that *multivariable linear regression* is a hyperplane
 - ▶ This hyperplane represents the expected value of a continuous outcome, Y , estimated via least squares for a set of predictors
- ▶ Support vector machines seek a *separating hyperplane*
 - ▶ $f(\mathbf{X}) > 0$ for one class, and $f(\mathbf{X}) < 0$ for the other class

Separating Hyperplanes

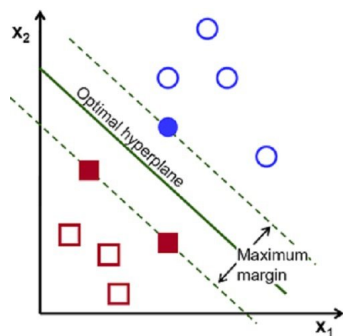
Consider two features, X_1 and X_2 , and a binary outcome. It might be possible to draw several separating hyperplanes:



Which of these hyperplanes is the better classifier?

The Maximum Margin Classifier

A *hard margin* SVM finds the “maximum margin” hyperplane:



This plane represents the “widest street” between classes, and it is characterized by “support vectors”, or training data-points that would change this hyperplane if removed

The Maximum Margin Classifier

- ▶ Consider the constraint: $\sum_{j=1}^p \beta_j^2 = 1$, which normalizes how our hyperplane is defined
 - ▶ This won't impact the direction of the plane, as $\{\beta_1 = 1, \beta_2 = 1\}$ and $\{\beta_1 = 3, \beta_2 = 3\}$ have the same orientation
- ▶ The SVM is defined by $\beta_1, \dots, \beta_p$ which maximize M in the expression: $y_i(\beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_p x_{i,p}) \geq M \quad \forall i = 1, \dots, n$
 - ▶ Here the binary outcome, y_i , is encoded as $+1$ or -1 , so the left side of this expression is the distance from the current hyperplane to the i^{th} data-point

The Maximum Margin Classifier

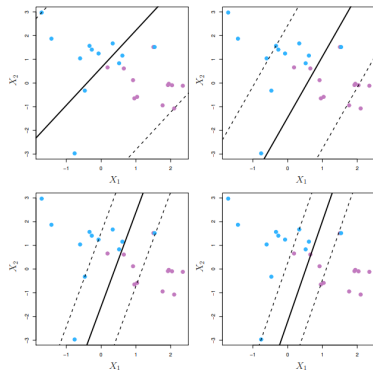
- ▶ The coefficients defining the SVM classifier can be found using the Lagrangian multiplier method
 - ▶ We will not cover this method in this course (as SVMs are the only classifier we'll study that use it)
- ▶ If you're interested in the mathematical details, I recommend Robert Berwick's (of MIT) "An Idiot's guide to support vector machines"

Soft Margin Classifiers

- ▶ For non-separable data, we can relax the maximum margin approach to find a *soft margin classifier*.
- ▶ Now we aim to find $\beta_1, \dots, \beta_p$ that maximize M where $y_i(\beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_p x_{i,p}) \geq M(1 - \epsilon_i)$
 - ▶ Subject to $\epsilon \geq 0$ and $\sum_{i=1}^n \epsilon_i < s$
 - ▶ $\epsilon_i = 0$ if a point is on the correct side of the margin
 - ▶ $0 < \epsilon_i < 1$ if a point is within the margin
 - ▶ $\epsilon_i > 1$ if a point is on the wrong side of the margin
 - ▶ s controls the total amount of “slack” that is allowed, with larger values allowing for more “slack”
 - ▶ As s decreases the tolerance for data-points being on the wrong side of the hyperplane diminishes

Soft Margin Classifiers

As s decreases (left to right), the margin M decreases:

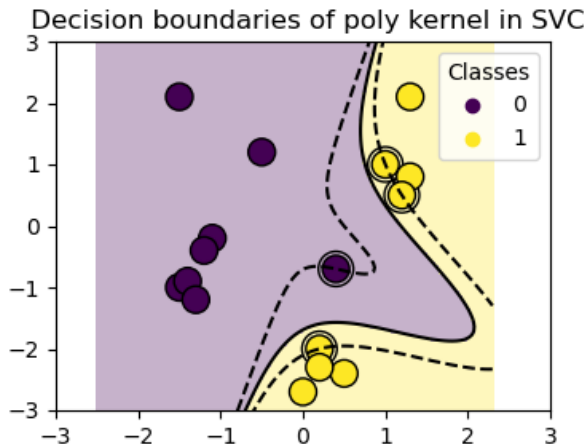


A larger s yields a more stable classifier, so the bias-variance trade-off can be manipulated via the value of s .

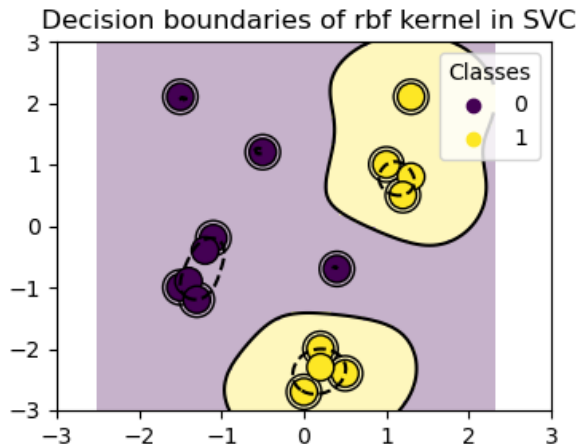
Feature Expansion and Kernel Functions

- ▶ Consider the features: $\{X_1, X_2\}$, and recall that the SVM classifier finds a decision boundary (separating hyperplane) of the form $\beta_0 + \beta_1 X_1 + \beta_2 X_2$
- ▶ We could apply transformations to create a new set of features: $\{X_1, X_2, X_1^2, X_2^2, X_1 X_2\}$
 - ▶ Now the decision boundary would have the form:
 $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2$
- ▶ This corresponds to a *non-linear boundary* in the *original feature space*
 - ▶ *Kernel functions* allow for computationally efficient mappings of the original features to higher dimensions for the purpose of finding a non-linear decision boundary

Polynomial Kernel ($d = 3, \gamma = 2$)

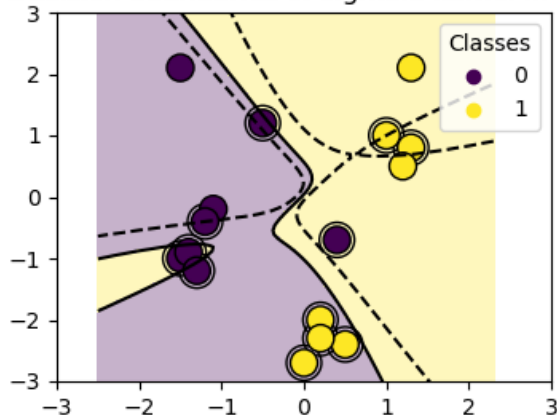


Radial Basis Function Kernel



Sigmoid Kernel

Decision boundaries of sigmoid kernel in SVC



Practical guidance

- ▶ SVMs treat each feature equally, so standardization is an important data preparation step
- ▶ The kernel function (type of feature expansion) and “slack” parameter can be tuned via cross-validation to achieve optimal classification performance
 - ▶ `sklearn` represents “slack” using a parameter `C` that is inversely proportional to what we called `s`
 - ▶ Other hyperparameters affiliated with certain kernel functions, such as `\gamma` can also be tuned in this manner
- ▶ Support vector regression is also implemented in `sklearn`, the SVM lab will briefly cover this method
 - ▶ SVMs also have been generalized to multi-class tasks, and use a one-vs-one scheme in `sklearn`